

Technical Document #229: Software Engineering - Comprehensive Overview

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Domain-driven design (DDD) models software around business domains. It improves communication between technical and business stakeholders.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

Microservices architecture structures applications as independently deployable services. Each service owns its data and business logic.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Continuous deployment (CD) automatically deploys code changes to production.
- Test-driven development (TDD) writes tests before writing code.
- Microservices architecture structures applications as independently deployable services.